

Sentinel Secure RAG Framework

Overview

Sentinel is a production-grade Secure Retrieval-Augmented Generation (RAG) framework designed to protect enterprise AI systems against prompt injection, jailbreak attacks, sensitive information disclosure, and malicious document ingestion.

The framework consists of several independent but collaborative components.

Architecture

The architecture contains the following major subsystems.

Authentication Layer

The authentication layer validates every incoming user before allowing access to protected resources.

Features include:

- Role Based Access Control (RBAC)
- Session validation
- Authentication token verification
- Audit logging

Input Security Agent

The Input Security Agent validates every user request before it reaches the language model.

Responsibilities include:

- Prompt Injection Detection
- Jailbreak Detection
- PII Detection
- Malicious Instruction Detection
- Similarity Detection
- Policy Evaluation
- Risk Scoring

Knowledge Base

The Knowledge Base consists of:

- Markdown Loader
- Recursive Chunker
- Sentence Transformer Embedder
- ChromaDB Vector Store
- Semantic Retriever

Documents are indexed only after passing document security validation.

Assistant Agent

The Assistant Agent coordinates:

- Secure Retrieval
- Context Construction
- Prompt Construction
- LLM Invocation

The assistant never bypasses the security layer.

Output Validation Agent

The Output Validation Agent verifies generated responses before returning them to the user.

Responsibilities include:

- Sensitive information detection
- Hallucination checks
- Policy validation
- Output filtering

Security Principles

Sentinel follows the following principles.

1. Least Privilege
2. Zero Trust
3. Defense in Depth
4. Secure by Default
5. Principle of Explicit Validation

Conclusion

Sentinel demonstrates how multiple security agents can protect modern enterprise AI systems while preserving Retrieval-Augmented Generation capabilities.